STATISTICAL ANALYSIS REPORT

Real Estate Valuation Dataset

Exploratory Data Analysis, Descriptive Statistics, Distributional Assessment, Outlier Detection, Correlation and Multicollinearity Diagnostics

Project: Ridge Regression and Multicollinearity

Programme: M.Tech. Modeling and Simulation

Institution: Defence Institute of Advanced Technology (DIAT), Pune

Prepared from the eight Jupyter notebooks supplied for this analysis

Analysis dataset: 414 observations and 7 original variables

Executive Summary

This report documents the statistical examination of the UCI Real Estate Valuation dataset as implemented in the supplied Jupyter notebooks. The analysis covers data integrity checks, conversion of the transaction-date field into year and month components, descriptive statistics, distributional patterns, IQR-based outlier screening, predictor–response relationships, Pearson correlation analysis, and multicollinearity diagnostics.

The dataset contains 414 observations. The response variable is house price of unit area. Among the predictors, distance to the nearest MRT station has the strongest absolute bivariate correlation with the response ($r = -0.6736$), followed by number of convenience stores ($r = 0.5710$), latitude ($r = 0.5463$), and longitude ($r = 0.5233$). House age has a weaker negative association ($r = -0.2106$), while transaction year has a very weak positive association ($r = 0.0815$).

The distance-to-MRT variable is strongly right-skewed (skewness 1.8888) and has 37 observations outside the 1.5×IQR upper fence. Latitude, longitude and the response also contain IQR-flagged observations. Such flags identify values that warrant examination; they do not establish that observations are erroneous or should be removed.

The raw predictor matrix has a very large condition number (237,426), while the standardized predictor matrix has a condition number of 4.2097. This substantial change indicates that scale and centering contribute heavily to the raw numerical conditioning. The standardized VIF results range from 1.0055 to 4.2938, with the highest value for distance to MRT. The diagnostics indicate some predictor association, especially between MRT distance and longitude, but do not support describing the standardized design as severely globally collinear.

Scope note: This report documents the statistical analysis already present in the notebooks. It does not claim that OLS, QR, SVD, ridge regression, ridge trace analysis, or cross-validation have been completed; those are subsequent modelling tasks.

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1. Introduction and Objectives

The purpose of this stage is to understand the structure and statistical properties of the real-estate dataset before fitting regression models. In a regression study, preliminary analysis is useful for identifying data-quality issues, understanding variable scales and distributions, examining linear associations, and assessing whether relationships among predictors may affect coefficient estimation.

The work documented here is an exploratory and diagnostic stage. It establishes an empirical baseline for the later comparison of ordinary least squares (OLS) solution methods and the investigation of ridge regularization.

1.1 Objectives

• Verify the integrity and numerical usability of the source data.

• Describe the central tendency, dispersion, skewness and kurtosis of the variables.

• Inspect the distributions of the numerical variables.

• Flag potential outliers using the 1.5×IQR rule without automatically deleting them.

• Examine bivariate relationships between each predictor and house price.

• Quantify linear associations among predictors and between predictors and the response.

• Assess numerical conditioning and predictor collinearity using condition numbers and VIF.

1.2 Dataset source and analytical unit

The first notebook retrieves the Real Estate Valuation dataset through the UCI Machine Learning Repository interface (fetch_ucirepo, dataset identifier 477). The analytical unit is one real-estate transaction/property record. The original table has 414 rows and seven columns.

2. Dataset Description and Variables

Symbol / field	Role	Description	Observed scale / units
X1 transaction date	Predictor	Transaction date encoded as a decimal year in the source dataset.	2012.667–2013.583
X2 house age	Predictor	Age of the house.	Years
X3 distance to nearest MRT station	Predictor	Distance to the nearest Mass Rapid Transit station.	Dataset distance units
X4 number of convenience stores	Predictor	Number of nearby convenience stores.	Count (0–10)
X5 latitude	Predictor	Geographic latitude of the property.	Decimal degrees
X6 longitude	Predictor	Geographic longitude of the property.	Decimal degrees
Y house price of unit area	Response	House price per unit area.	Dataset response units

The date-preparation notebook derives a transaction year and a month name from the decimal transaction-date field. The month-number conversion in the supplied notebook uses the fractional year multiplied by 12, rounded, and shifted to calendar numbering with +1. The user has confirmed that this conversion has been validated. The derived year and month are retained as separate descriptive fields; month is categorical and is excluded from numerical correlation and regression-predictor calculations described in the EDA notebooks.

Response variable: Y house price of unit area. **Predictor set used in numerical diagnostics:** transaction year, house age, MRT distance, convenience-store count, latitude and longitude.

3. Data Integrity and Preprocessing

The source-data inspection notebook checks the table dimensions and column types, missing values, empty strings, duplicate rows, numeric convertibility, infinite values, unique-value counts, descriptive summaries and the raw

correlation matrix.

Check	Observed result	Interpretation
Dataset dimensions	414 rows × 7 columns	All 414 records are represented in the analysis.
Missing values	0	No missing cells were reported.
Duplicate rows	0	No exact duplicate rows were reported.
Infinite numeric values	0	No positive or negative infinity values were reported.
Numeric conversion	No invalid conversions reported	The source variables were numerically parseable.
Type consistency	6 float columns; 1 integer column	Convenience-store count is integer-valued; remaining source fields are floating point.

The source data therefore passed the checks implemented in the notebook. The transaction-date conversion creates a year field and a month-name field for subsequent summaries. No missing-value imputation, row deletion, or outlier removal is documented in the supplied EDA notebooks.

3.1 Preprocessing boundary

The analysis uses the prepared CSV for descriptive statistics, distributions, outlier screening, feature relationships, correlations and multicollinearity diagnostics. The date conversion is treated as validated as confirmed by the user. This report preserves the reported notebook outputs rather than silently changing the transformation or recomputing the analysis under a different date convention.

A useful distinction for the later modelling stage is that preprocessing for regression may include choosing a consistent predictor matrix, defining an intercept, and optionally scaling predictors. These are modelling-design choices and do not imply that the current data-cleaning stage is incomplete.

4. Descriptive Statistical Analysis

Descriptive statistics summarize the location and spread of each numerical variable. The notebook reports count, mean, standard deviation, minimum, quartiles, median and maximum, together with variance, skewness and kurtosis. The month field is summarized separately as a categorical variable.

Variable	Mean	Std. dev.	Median	Minimum	Maximum
Transaction year	2012.6957	0.4607	2013.0000	2012	2013
House age	17.7126	11.3925	16.1000	0.0	43.8
MRT distance	1083.8857	1262.1096	492.2313	23.3828	6488.0210
Convenience stores	4.0942	2.9456	4.0000	0	10
Latitude	24.9690	0.0124	24.9711	24.93207	25.01459
Longitude	121.5334	0.0153	121.5386	121.47353	121.56627
House price/unit area	37.9802	13.6065	38.4500	7.6	117.5

4.1 Quartile structure

Variable	Q1 (25%)	Median (50%)	Q3 (75%)
Transaction year	2012	2013	2013
House age	9.025	16.100	28.150
MRT distance	289.3248	492.2313	1454.2790
Convenience stores	1	4	6
Latitude	24.963000	24.971100	24.977455
Longitude	121.528085	121.538630	121.543305

Variable	Q1 (25%)	Median (50%)	Q3 (75%)
House price/unit area	27.7	38.45	46.6

MRT distance has a mean (1,083.89) substantially above its median (492.23), suggesting a long upper tail. The standard deviation (1,262.11) also exceeds the mean, indicating substantial dispersion relative to its central value. House price has a mean of 37.98 and median of 38.45, with a standard deviation of 13.61 and a maximum of 117.5.

The convenience-store variable is discrete and bounded between zero and ten. Latitude and longitude have narrow numerical ranges because they represent geographic coordinates; their small standard deviations should not be interpreted as low importance. Variable scales differ substantially, which is relevant when assessing numerical conditioning.

4.2 Skewness and kurtosis

Variable	Skewness	Excess kurtosis	Reading
Transaction year	−0.8535	−1.2777	Left-skewed; limited to two calendar years.
House age	0.3829	−0.8771	Mild right skew; flatter than a normal distribution by this measure.
MRT distance	1.8888	3.2079	Strong right skew and heavier tails.
Convenience stores	0.1546	−1.0658	Approximately symmetric discrete count distribution.
Latitude	−0.4386	0.2691	Mild left skew.
Longitude	−1.2196	1.2018	Negative skew with positive excess kurtosis.
House price/unit area	0.5999	2.1791	Moderate right skew and heavier upper tail.

Pandas reports kurtosis as excess kurtosis, for which a normal distribution has a reference value of zero. Positive values indicate heavier tails or greater tail concentration relative to the normal reference; negative values indicate lighter tails. These are descriptive shape measures and do not, by themselves, establish whether a variable is normally distributed.

4.3 Categorical summaries

The year-frequency output reports 288 transactions in 2013 and 126 in 2012. The month-frequency output saved in the notebook is reproduced below.

Month	Count	Month	Count
June	58	July	47
February	46	December	38
April	32	November	31
September	30	May	29
January	28	October	27
March	25	August	23

June is the most frequent month in the saved summary (58 observations), while August is the least frequent (23). These counts describe the distribution of the prepared categorical month field; they do not imply a causal seasonal effect on price.

5. Distributional Analysis

The distribution-analysis notebook plots histograms for the six numerical variables other than transaction year. The plots are used to inspect concentration, spread, asymmetry, discreteness and possible tail behaviour.

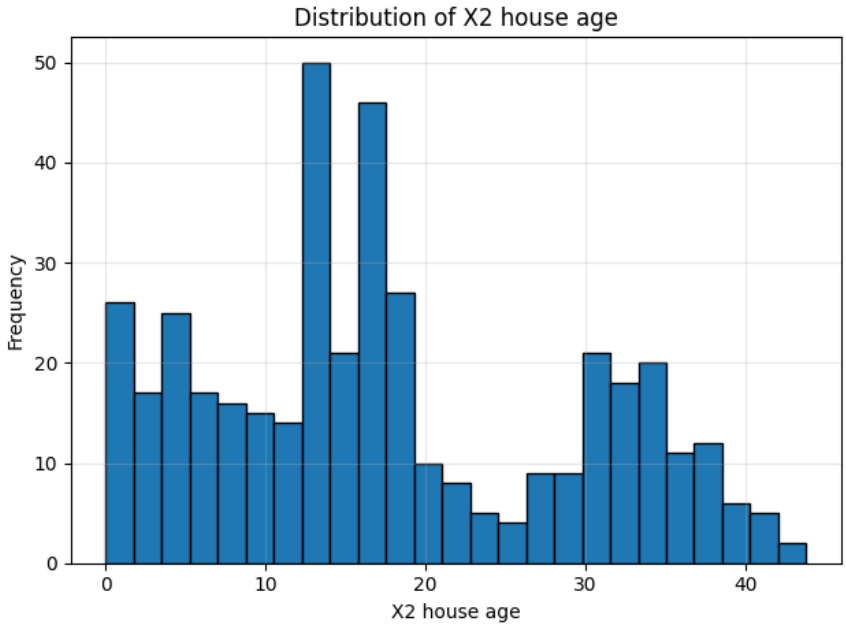


Figure 1. Distribution of house age.

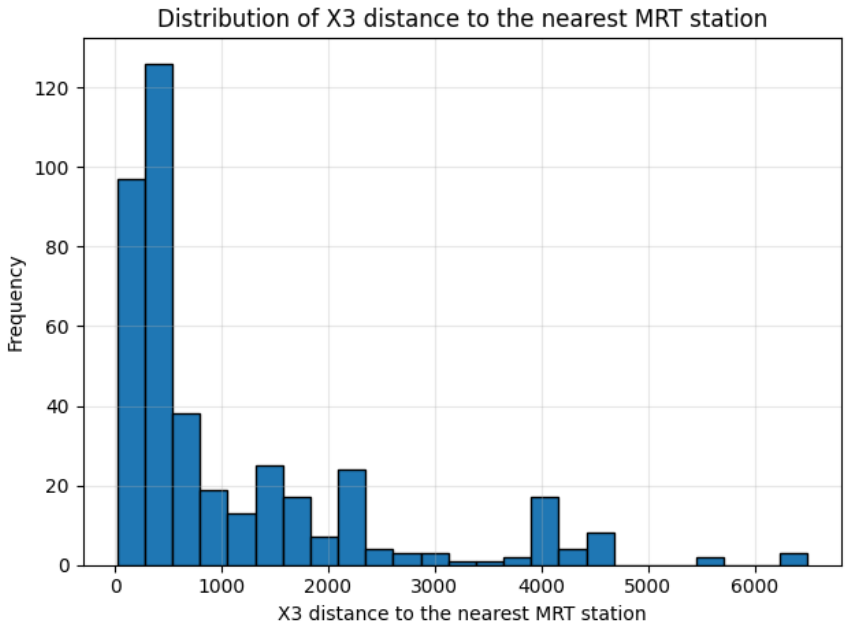


Figure 2. Distribution of distance to the nearest MRT station.

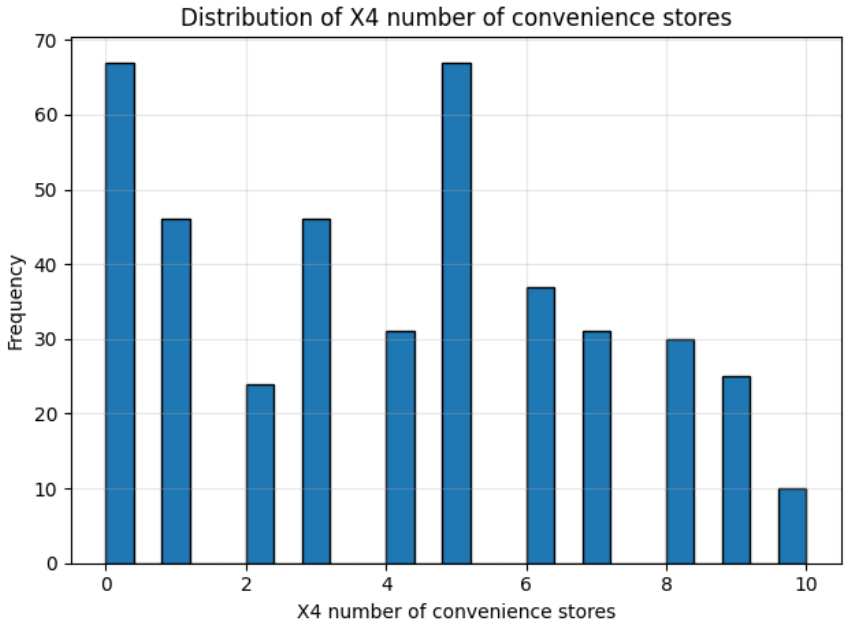


Figure 3. Distribution of convenience-store count.

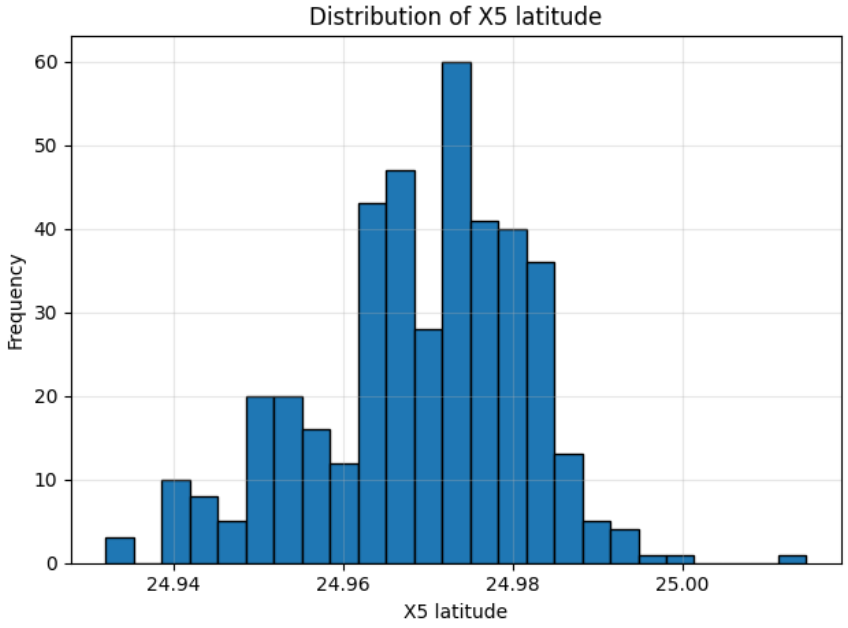


Figure 4. Distribution of latitude.

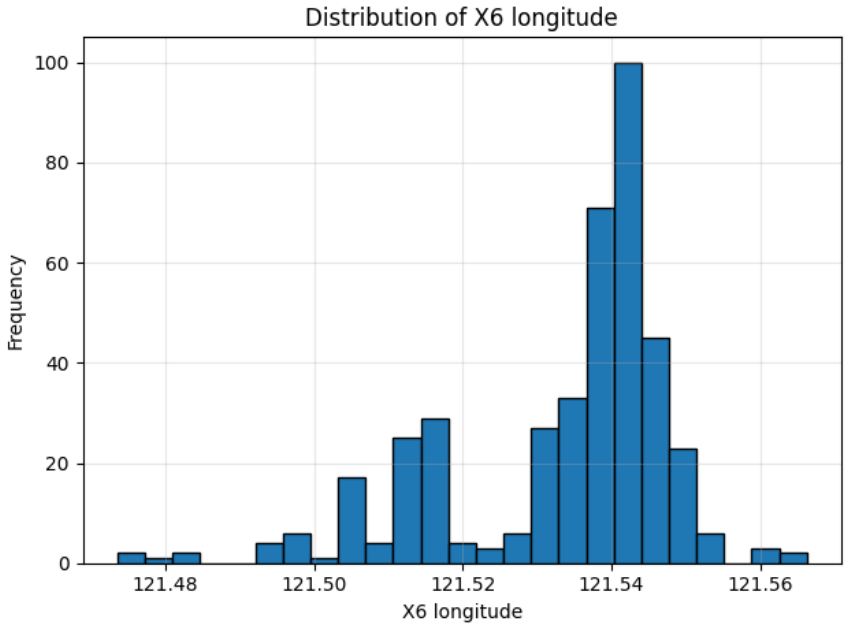


Figure 5. Distribution of longitude.

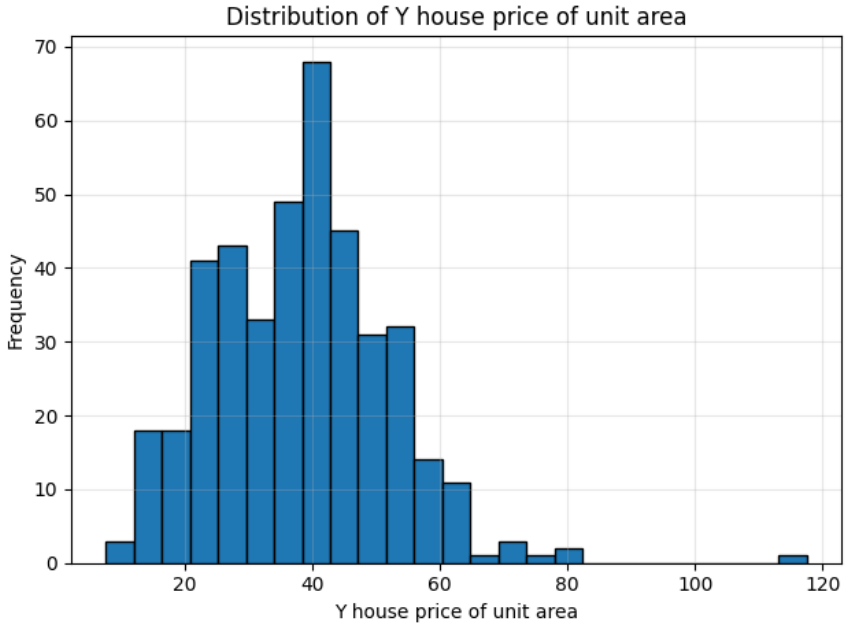


Figure 6. Distribution of house price per unit area.

5.1 Interpretation of the histograms

• **House age:** The observations are spread across the 0–43.8 year range and show uneven concentrations rather than a single sharply defined peak.

• **MRT distance:** The distribution is strongly right-skewed. Most observations occur at comparatively shorter distances, with a long tail extending to 6,488.021.

• **Convenience-store count:** The variable is discrete, taking integer values from 0 to 10. Its histogram should be interpreted as a count distribution, not a continuous measurement.

• **Latitude:** Values are tightly concentrated within a narrow geographic band, with a modest lower-side tail.

• **Longitude:** The distribution is concentrated around 121.54, with a longer lower-side tail, consistent with its negative skewness.

• **House price:** Most observations are concentrated in the central range, with a right tail extending to 117.5.

Histogram shapes are descriptive. They do not establish normality of regression errors, which is a separate model-diagnostic question to be considered after a regression model is fitted.

6. Outlier Analysis Using the IQR Rule

Potential outliers were screened using the interquartile range (IQR). For a variable with first quartile Q1 and third quartile Q3, IQR = Q3 − Q1. The lower and upper screening fences are Q1 − 1.5(IQR) and Q3 + 1.5(IQR), respectively. Values outside these fences are flagged.

IQR = Q3 − Q1
Lower fence = Q1 − 1.5 × IQR
Upper fence = Q3 + 1.5 × IQR

Variable	Lower fence	Upper fence	Flagged count
Transaction year	2010.5000	2014.5000	0
House age	−19.6625	56.8375	0
MRT distance	−1458.1065	3201.7103	37
Convenience stores	−6.5	13.5	0
Latitude	24.9413175	24.9991375	8
Longitude	121.505255	121.566135	35
House price/unit area	−0.65	74.95	3

6.1 Interpretation

MRT distance has 37 IQR-flagged observations, all on the upper side because the lower fence is negative while observed distances are non-negative. The target has three flagged observations, all on the upper side because its lower fence is below the observed minimum. Latitude has eight flags and longitude has 35. No values are flagged for year, house age or convenience-store count.

Geographic-coordinate flags may reflect genuine spatial variation or properties located toward the edges of the sampled region. High MRT distances may represent valid remote properties. High house-price values may be genuine premium observations. The IQR rule is a screening heuristic and does not distinguish data-entry errors from rare but valid observations.

Decision: The notebook reports outlier counts and fences; it does not document deleting or winsorizing these observations. Retaining them at this exploratory stage avoids changing the empirical dataset without a substantive reason.

For later regression diagnostics, it may be useful to inspect the flagged rows and assess leverage, residuals and influence after fitting a model. Those are model-dependent diagnostics and are not equivalent to univariate IQR screening.

7. Predictor–Response Relationships

The feature-relationships notebook generates a scatterplot for each numerical predictor against house price per unit area, adds a simple linear trendline, and reports the Pearson correlation coefficient. These are bivariate relationships: each predictor is considered individually, without controlling for the other predictors.

Predictor	Pearson r with price	Bivariate interpretation
Transaction year	0.082	Very weak positive association.
House age	−0.211	Weak negative association.
MRT distance	−0.674	Strongest absolute association; price tends to be lower at greater MRT distance.
Convenience stores	0.571	Moderate positive association.

Predictor	Pearson r with price	Bivariate interpretation
Latitude	0.546	Moderate positive association.
Longitude	0.523	Moderate positive association.

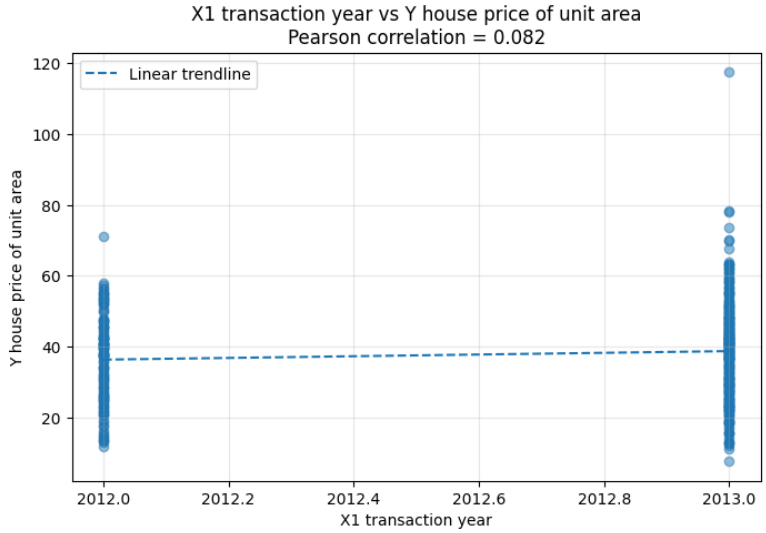


Figure 7. Transaction year versus house price per unit area.

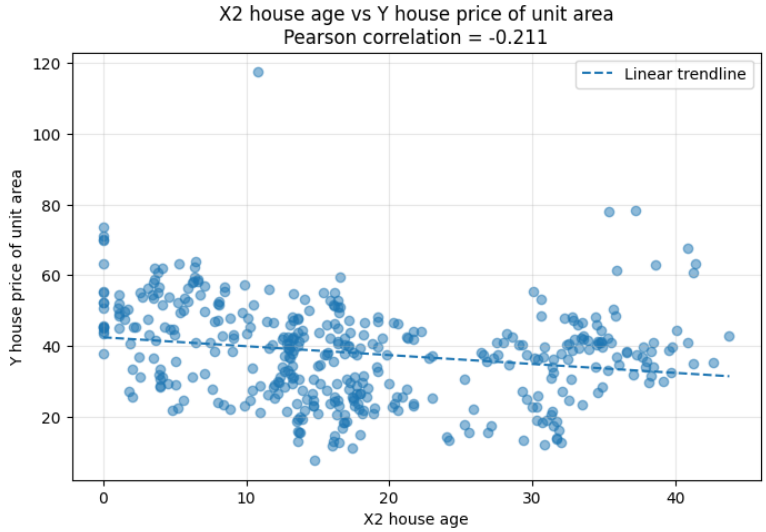


Figure 8. House age versus house price per unit area.

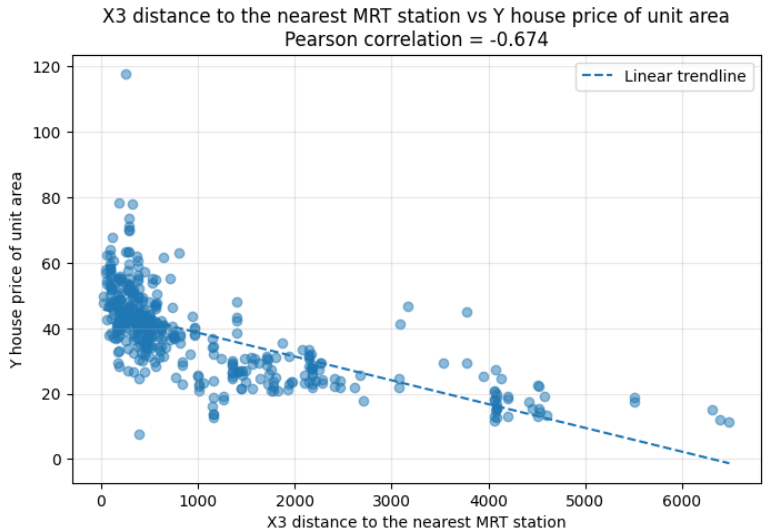


Figure 9. MRT distance versus house price per unit area.

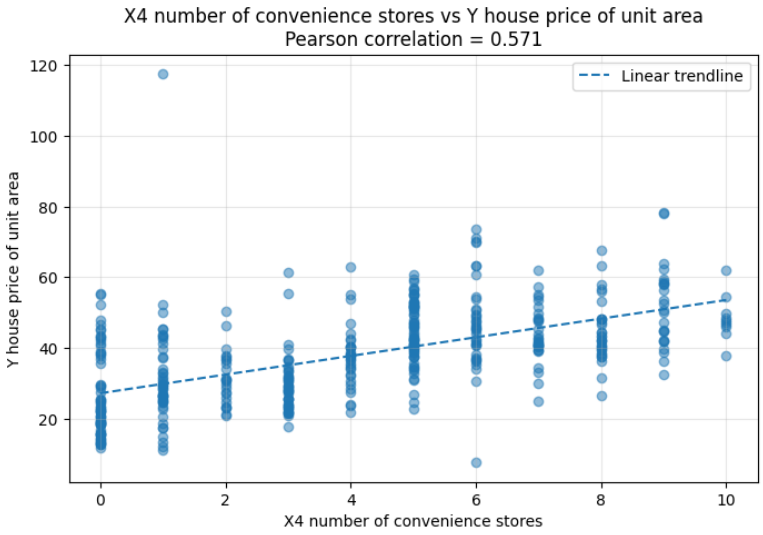


Figure 10. Convenience-store count versus house price per unit area.

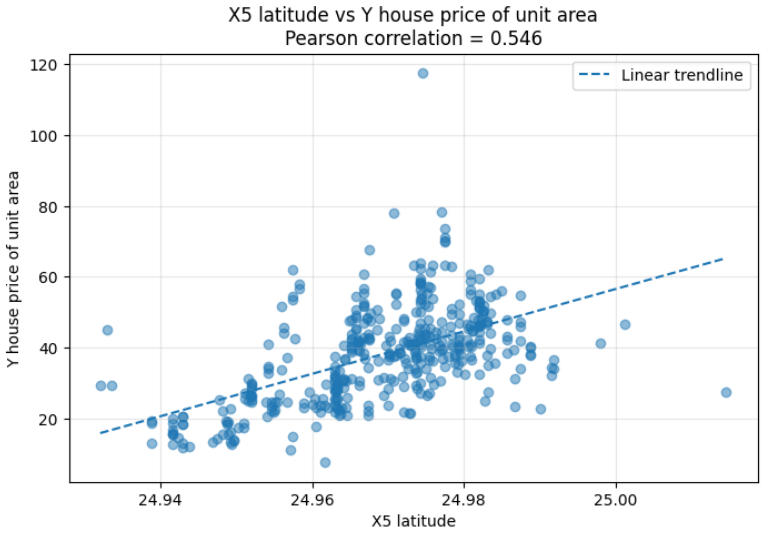


Figure 11. Latitude versus house price per unit area.

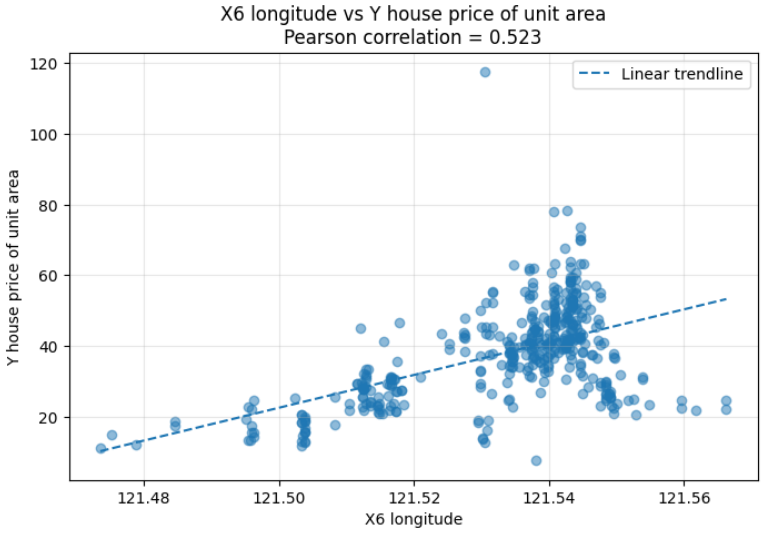


Figure 12. Longitude versus house price per unit area.

7.1 Interpretation and caution

MRT distance has the largest absolute bivariate correlation with the response (|r| ≈ 0.674), and its negative sign indicates that greater distance is associated with lower unit-area price in this sample. Convenience-store count, latitude and longitude show moderate positive associations. House age has a weak negative association. Transaction year has a very weak association; because it takes only two distinct year values in the prepared data, it should not be interpreted as a smooth temporal trend.

The scatterplots and Pearson coefficients do not establish causality. They also do not reveal adjusted or partial effects. A predictor can have a different coefficient or even a different sign in a multiple regression because predictors may be correlated with one another.

8. Correlation Analysis

The correlation-analysis notebook computes the Pearson product-moment correlation matrix for the numerical variables, visualizes it as a heatmap, ranks predictor–target correlations by absolute magnitude, and inspects pairwise predictor correlations. Pearson's r measures the strength and direction of a linear association and ranges from –1 to +1.

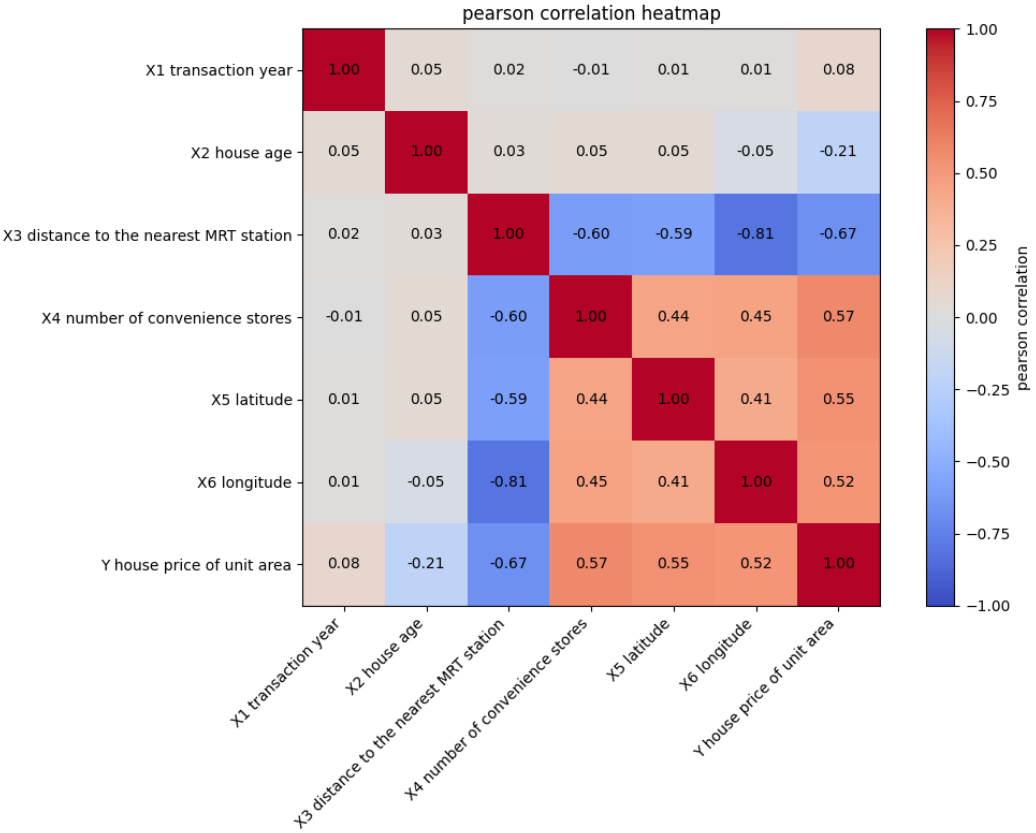


Figure 13. Pearson correlation heatmap for numerical variables.

8.1 Predictor–target correlations

Predictor	Pearson r with price	Bivariate interpretation
Transaction year	0.082	Very weak positive association.
House age	–0.211	Weak negative association.
MRT distance	–0.674	Strongest absolute association; price tends to be lower at greater MRT distance.
Convenience stores	0.571	Moderate positive association.
Latitude	0.546	Moderate positive association.

Predictor	Pearson r with price	Bivariate interpretation
Longitude	0.523	Moderate positive association.

8.2 Predictor–predictor correlations

Predictor pair	Pearson r	Absolute r
MRT distance – longitude	–0.806317	0.806317
MRT distance – convenience stores	–0.602519	0.602519
MRT distance – latitude	–0.591067	0.591067
Convenience stores – longitude	0.449099	0.449099
Convenience stores – latitude	0.444143	0.444143
Latitude – longitude	0.412924	0.412924

Using the notebook's selected threshold $|r| \geq 0.7$, only the pair consisting of MRT distance and longitude crosses the threshold. Their correlation is strongly negative (–0.8063). Other notable associations include moderate negative correlations between MRT distance and convenience-store count (–0.6025) and between MRT distance and latitude (–0.5911).

Pairwise correlation screening is useful but incomplete. A predictor may be approximately explained by a combination of several other predictors even when no individual pairwise correlation is extreme. This is why the subsequent VIF and condition-number analysis is relevant.

9. Multicollinearity Diagnostics

The multicollinearity notebook uses the six numerical predictors and excludes the response. Constant columns are removed if present. It computes the 2-norm condition number of the raw predictor matrix, standardizes the predictors using StandardScaler, recomputes condition numbers, and calculates variance inflation factors (VIF) on the standardized predictor matrix.

9.1 Condition number

Matrix	Condition number
Raw predictor matrix X	237,426.0354
Standardized predictor matrix X_scaled	4.2097
Raw cross-product matrix X■X	56,371,122,199.4602
Standardized cross-product matrix X_scaled■X_scaled	17.7214

The condition number of a matrix is the ratio of its largest singular value to its smallest singular value (for the 2-norm). A large condition number indicates that some directions in the matrix are much less well determined than others and that numerical solutions can be sensitive to perturbations.

The raw predictor matrix combines variables measured on very different scales and with large offsets: for example, transaction year is around 2012, longitude is around 121, MRT distance is around 1,084 on average, and latitude is around 25. The raw condition number is therefore influenced by scale and centering as well as by relationships among predictors. StandardScaler centers each predictor and scales it to unit variance. After standardization, the condition number falls to 4.2097, indicating much better numerical conditioning.

> For a full-column-rank matrix under the 2-norm, $\kappa_■(X■X) = [\kappa_■(X)]^2$. The reported standardized values are consistent with this identity: $4.2097^2 \approx 17.7214$.

Consequently, the very large raw condition number should not be presented by itself as proof of severe multicollinearity. The comparison before and after standardization shows that variable scales and offsets are a major part of the raw conditioning problem.

9.2 Variance inflation factors

Predictor	VIF
MRT distance	4.2938
Longitude	2.9312
Convenience stores	1.6133
Latitude	1.5999
House age	1.0167
Transaction year	1.0055

VIF quantifies how much the variance of a coefficient estimator is inflated by linear relationships between one predictor and the remaining predictors, under the auxiliary-regression definition. MRT distance has the largest VIF (4.2938), followed by longitude (2.9312). The remaining predictors have VIFs close to one or below two.

Common rules of thumb sometimes treat VIF values around 5 or 10 as thresholds for concern, but these are not universal statistical laws. Under such conventions, the maximum VIF of 4.2938 suggests a moderate or elevated relationship rather than an unequivocally severe level. The result is consistent with the strong pairwise association between MRT distance and longitude.

9.3 Diagnostic conclusion

Taken together, the standardized condition number and VIF values suggest that the predictor matrix is reasonably well-conditioned after standardization, with some moderate collinearity concentrated around geographic variables, especially MRT distance and longitude. This is a more defensible conclusion than labelling the dataset as having severe global multicollinearity.

The VIF calculation in the notebook is performed on centered and scaled predictors without an explicitly appended constant column. Since StandardScaler centers the predictors, this convention is interpretable, but the modelling report should state the convention. When fitting OLS later, the intercept treatment and the design matrix used for condition-number calculations should be clearly specified.

10. Integrated Interpretation and Implications for Regression

The EDA provides several points that should guide the next modelling stage:

• **Predictor selection:** All six numerical predictors have been examined against the response. MRT distance, convenience-store count, latitude and longitude show the largest bivariate associations in absolute terms, but feature selection should not be based solely on pairwise correlation.

• **Scale:** Predictor units differ markedly. A raw design matrix may be poorly scaled numerically. A standardized-predictor analysis is useful for comparing numerical behaviour and coefficient magnitudes, provided the scaling transformation is handled consistently.

• **Multicollinearity:** MRT distance and longitude have a strong pairwise correlation, and MRT distance has the largest VIF. The relationship may affect coefficient uncertainty and stability in a multiple regression even if predictive performance remains acceptable.

• **Distribution and outliers:** MRT distance and house price have right tails. IQR flags should be examined, not automatically removed. Regression residual and influence diagnostics should be conducted after model fitting.

• **Model comparison:** Normal equations, QR and SVD should use the same observations, predictor matrix, target vector and intercept convention so that differences reflect the numerical solution method rather than inconsistent inputs.

10.1 Bridge to the OLS stage

The natural next step is to construct the design matrix X from transaction year, house age, MRT distance, convenience-store count, latitude and longitude, and the response vector y from house price per unit area. An intercept

column of ones should be included consistently in all three OLS solution approaches.

OLS objective: minimize ■y – Xβ■■²
Normal equations (full column rank): β■ = (X■X)■¹X■y

For numerical reliability, QR decomposition and SVD provide alternatives to explicitly forming and inverting X■X. The project can then compare estimated coefficients, fitted values, residual sums of squares and numerical stability. This report does not include those results because they are not present in the supplied notebooks.

11. Limitations and Methodological Notes

• **Exploratory, not causal:** Correlation and scatterplots describe association; they do not establish causal effects.

• **Bivariate relationships:** Feature–target correlations do not adjust for the other predictors. Multiple regression coefficients may differ.

• **Linear association:** Pearson correlation measures linear association and can miss nonlinear patterns.

• **IQR flags:** The 1.5×IQR rule is a screening convention. It does not classify observations as errors or justify automatic deletion.

• **Condition number dependence:** Condition numbers depend on scaling and on the exact design matrix, including whether an intercept is present.

• **VIF convention:** The notebook computes VIF on standardized predictors without an explicit constant. This should be documented when comparing with later software outputs.

• **Full-data scaling:** StandardScaler is fitted to the complete dataset in the diagnostic notebook. This is acceptable for exploratory diagnostics; for predictive evaluation with a train/test split or cross-validation, fit the scaler on each training fold only to avoid information leakage.

• **Model diagnostics pending:** Residual normality, heteroscedasticity, leverage, influence, predictive error and cross-validation have not been evaluated in this EDA stage.

12. Conclusion

The statistical analysis establishes that the 414-record real-estate dataset is complete under the data-integrity checks performed. The variables have heterogeneous scales and distributions. MRT distance is strongly right-skewed and has the strongest negative bivariate relationship with house price. Convenience-store count and the geographic coordinates have moderate positive bivariate associations with the response. House age has a weak negative association, while transaction year has a very weak positive association.

The IQR procedure flags 37 MRT-distance observations, eight latitude observations, 35 longitude observations and three house-price observations. These observations should be retained unless further contextual or model-based evidence indicates that they are erroneous or unduly influential.

The multicollinearity analysis identifies a strong pairwise association between MRT distance and longitude. However, after standardization, the predictor-matrix condition number is 4.2097 and the maximum VIF is 4.2938. The evidence therefore supports a nuanced conclusion: the data contain some moderate predictor collinearity, especially among geographic variables, but the standardized predictor matrix does not exhibit severe global ill-conditioning according to the diagnostics reported.

This EDA stage is complete as documented in the supplied notebooks. The project can now proceed to OLS estimation through normal equations, QR decomposition and SVD, followed by ridge regression, ridge-trace analysis and k-fold cross-validation as specified in the project work.

Appendix A. Notebook Traceability

Notebook	Role in this report
test.ipynb	Dataset retrieval, schema and integrity checks, original descriptive output and raw correlations.
updated_csv.ipynb	Conversion of decimal transaction date into year and month fields; user-confirmed validated conversion.
descriptive_statistics.ipynb	Numerical summaries, median, variance, standard deviation, skewness, kurtosis and categorical frequencies.
distribution_analysis.ipynb	Histograms for six numerical variables other than transaction year.
outlier_analysis.ipynb	IQR fences and outlier counts.
feature_relationships.ipynb	Predictor-versus-response scatterplots, linear trendlines and Pearson correlations.
correlation_analysis.ipynb	Pearson correlation matrix, heatmap, target-correlation ranking and predictor-pair screening.
multicollinearity.ipynb	Raw and standardized condition numbers, cross-product condition numbers and VIF.

Appendix B. Reproducibility and reporting conventions

Values in this report are transcribed from the saved notebook outputs supplied for review. Displayed values are rounded for readability; calculations in the notebooks retain their underlying numerical precision. Figures are embedded from the saved notebook outputs. The month-frequency table reproduces the saved output of descriptive_statistics.ipynb; the date conversion is retained as implemented and confirmed validated by the user.

The report intentionally distinguishes results already obtained from tasks still to be performed. No regression coefficients, R^2 values, residual diagnostics, ridge estimates, optimal λ value or cross-validation scores are reported because these are not present in the eight EDA notebooks.